

AYBUStore: Intelligent Campus E-Commerce Ecosystem

Supplementary Technical Report for TÜBİTAK 1001 Proposal

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Abstract

This report details the technical architecture, methodology, and requirement specifications for AYBUStore, a smart e-commerce platform developed for Ankara Yıldırım Beyazıt University. Utilizing a modern stack of Next.js 14 and Firebase, the project integrates AI-driven personalization and real-time inventory management to solve physical campus commerce inefficiencies.

1 Introduction & Problem Definition

The AYBUStore project addresses the lack of a centralized digital ecosystem for university-branded merchandise and department-specific materials. Existing solutions fail to provide real-time stock transparency, personalized apparel preview, or AI-integrated shopping assistance.

1.1 Scientific Excellence: Hyper-Local Closed-Loop Digital Ecosystem

As detailed by the research team (U. Dindaroğlu), the project conceptualizes a **Hyper-Local Closed-Loop Digital Ecosystem (HL-CLDE)** specifically tailored for a distributed university environment. Unlike standard e-commerce platforms, AYBUStore addresses the *Four-Campus Distribution Problem* where geographical fragmentation between Esenboğa, Etlik, Bilkent, and Çubuk campuses limits brand physical accessibility. By digitizing inventory and centralizing logistics, the platform targets a **30% increase in merchandise accessibility** for students. A core focus is the optimization of **Time-to-Interactive (TTI)**, aiming for a **1.2s benchmark** to ensure seamless user engagement across varied campus network conditions.

2 System Architecture

We adopt a **Component-Based Architecture** using Next.js App Router for optimal performance and SEO. The backend is powered by Firebase, providing a serverless real-time database and secure authentication.

2.1 Use Case Analysis

The following diagram illustrates the interactions between students, staff, and administrators within the ecosystem.

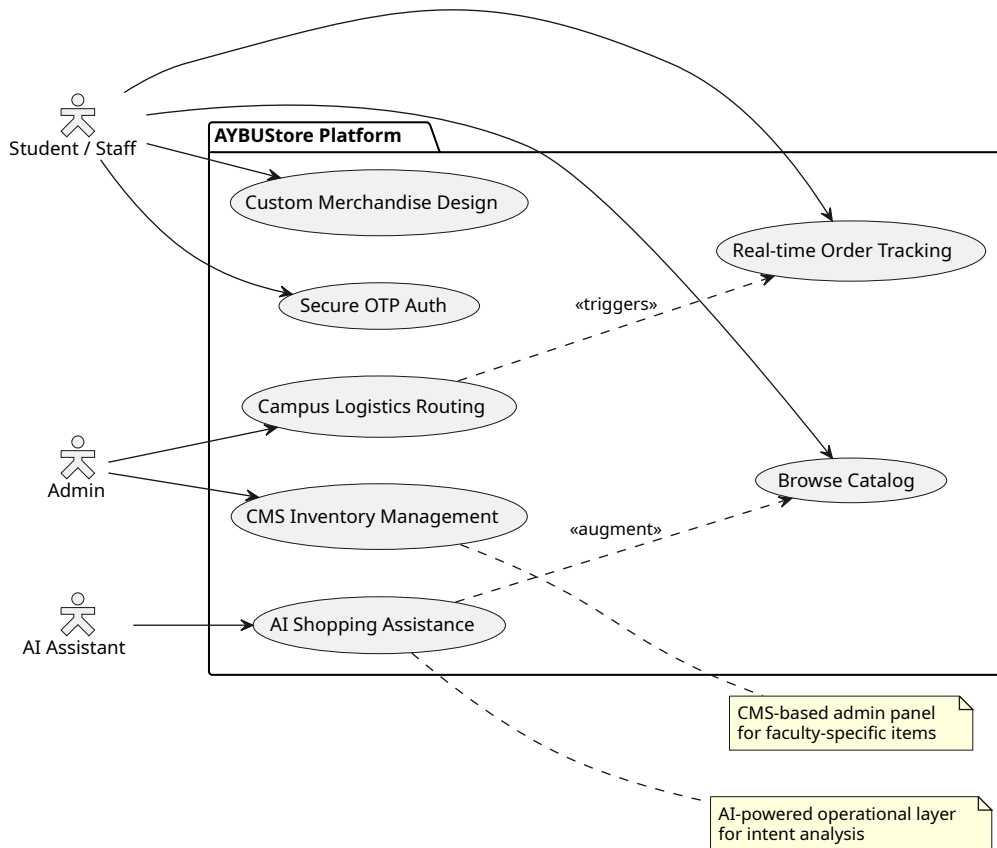


Figure 1: AYBUStore Use Case Diagram (Vector)

2.2 Domain Modeling

The system's data structure is designed with modularity in mind, ensuring scalability as new faculties are added.

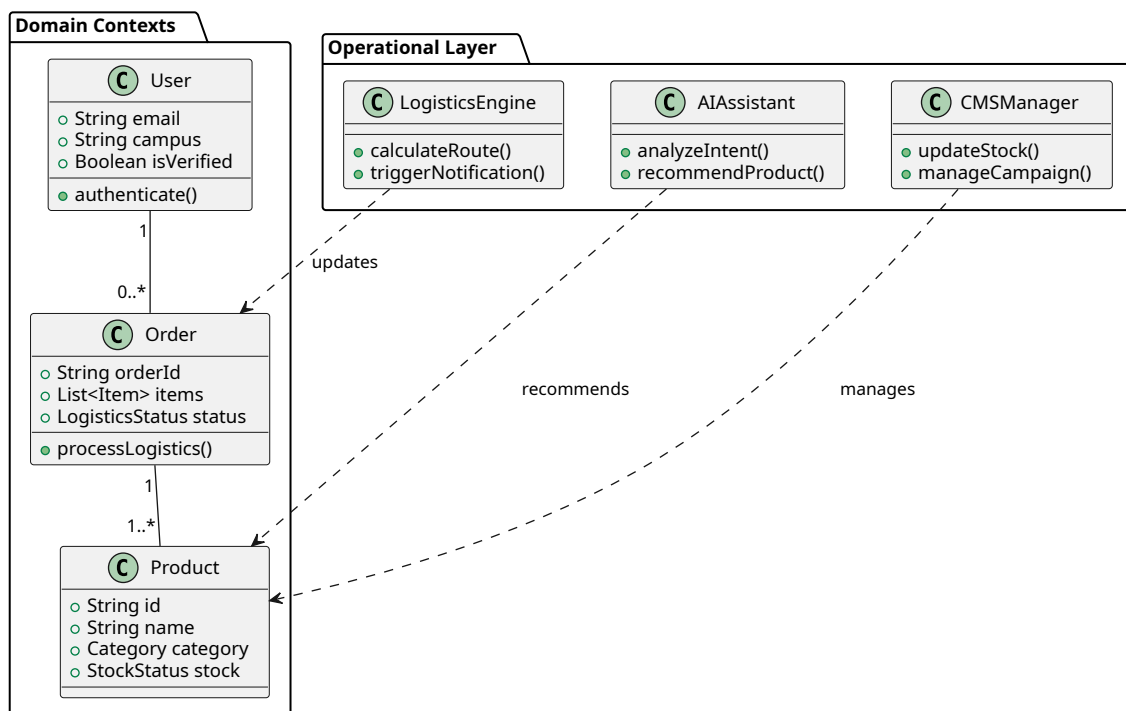


Figure 2: System Domain Class Diagram

3 Core Processes

3.1 Authentication & OTP Flow

Security is managed via a multi-stage authentication process, verified by a 6-digit one-time password (OTP) sent to university emails.

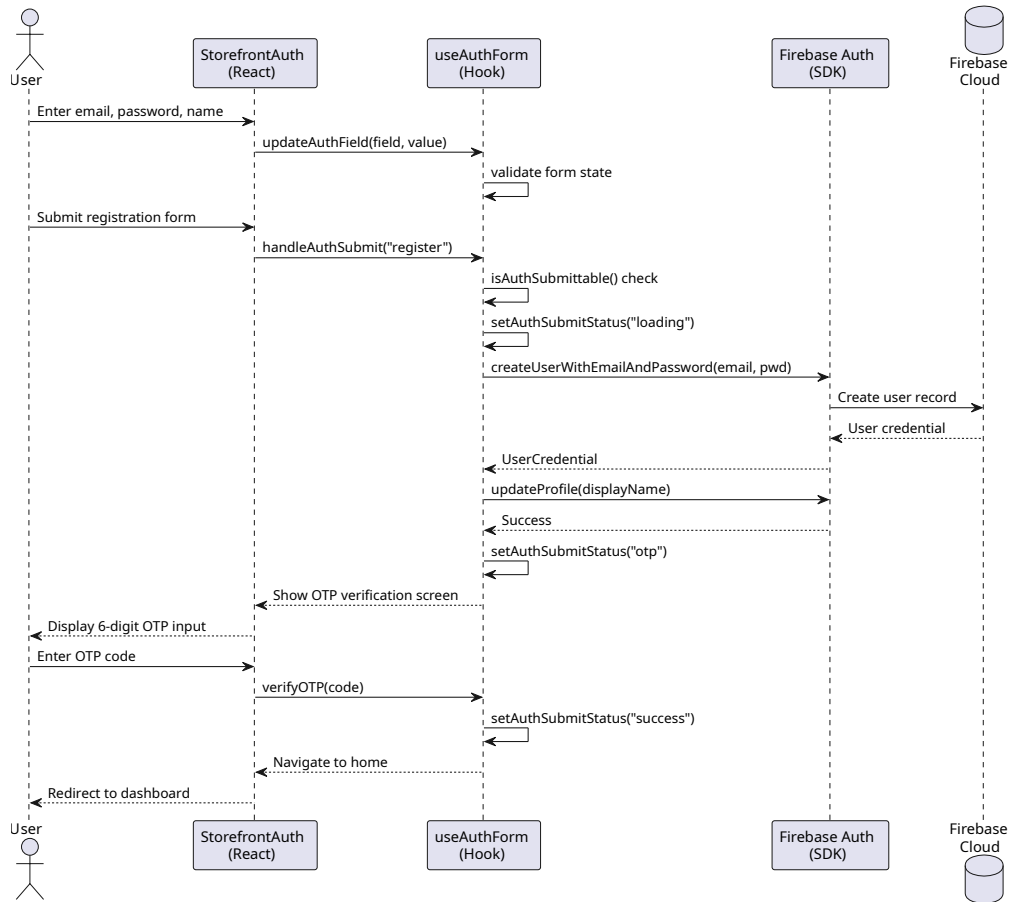


Figure 3: Registration and Security Verification Sequence

3.2 AI Personalization

The AI Shopping Assistant utilizes LLM-based intent analysis to provide context-aware product recommendations.

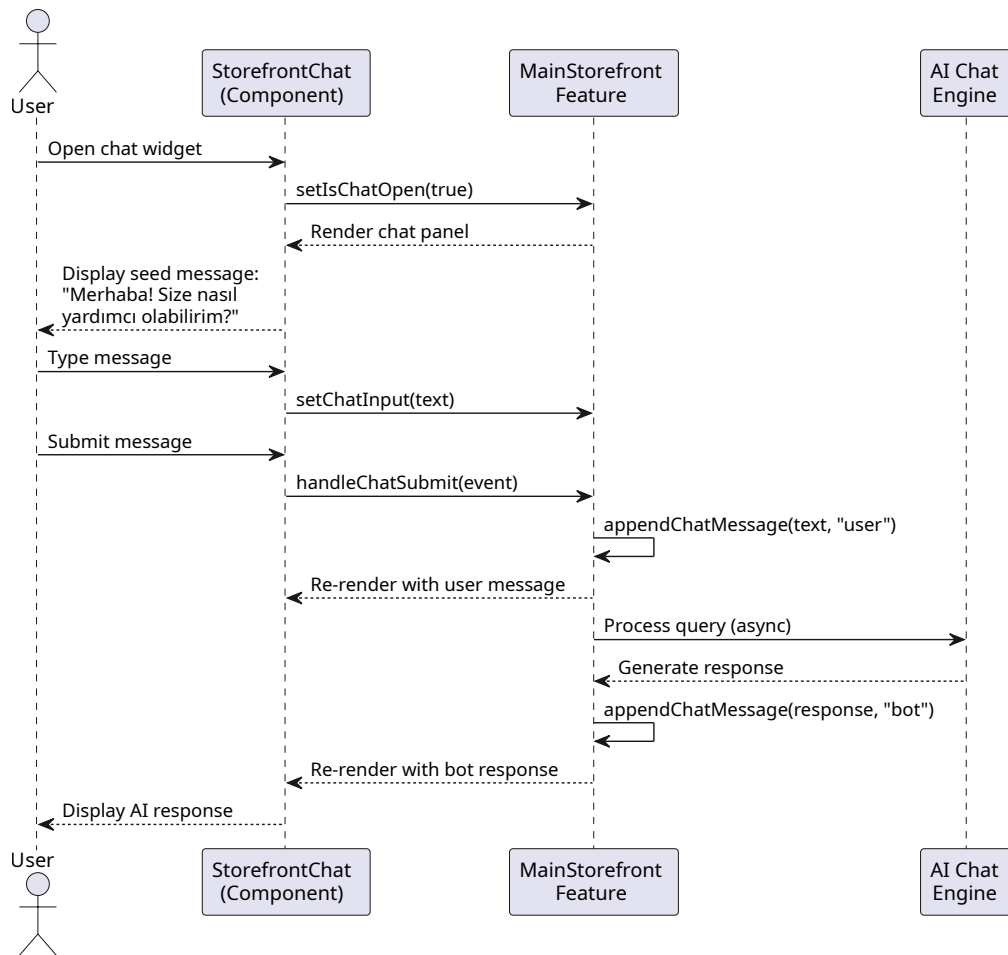


Figure 4: AI-Driven Assistant Interaction Flow

4 Operational Design

4.1 Shopping Lifecycle

The activity flow below describes the transition from browsing to custom design and final checkout.

Phase 4 Logistics & Notification Pipeline

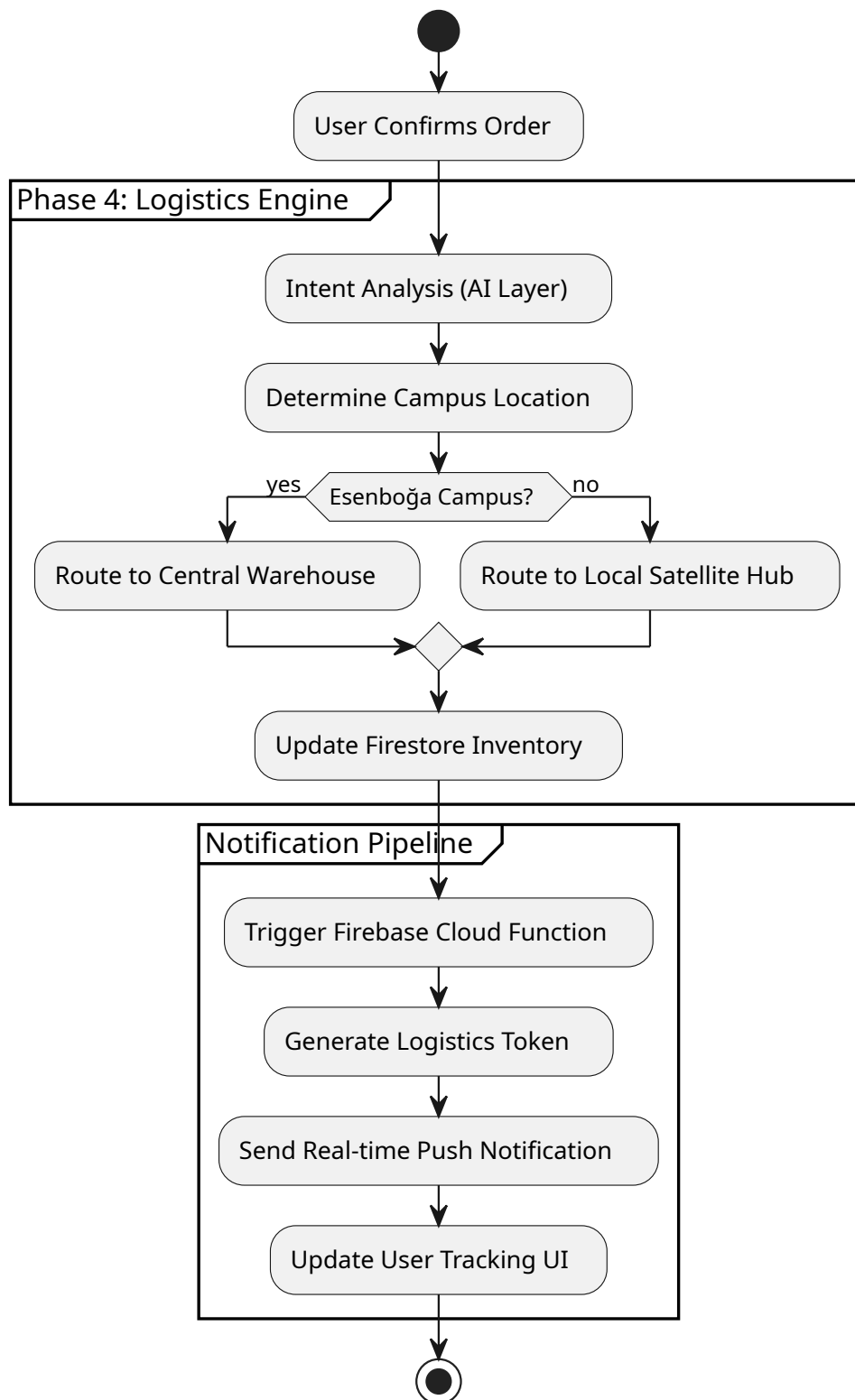


Figure 5: User Activity Lifecycle Diagram

5 Methodology & Software Requirement Specification

5.1 The 5-Phase Iterative Process

The development lifecycle follows a structured 5-phase iterative process as designed by A.S. Yilmaz:

- **Phase 1: Domain-Driven Design (DDD) Paradigm** - Mapping complex campus operations into bounded contexts (Authentication, Storefront, and Logistics).
- **Phase 2: Architectural Justification** - Selecting Node.js/Next.js over traditional Spring Boot or Python frameworks for superior non-blocking I/O and server-side rendering (SSR) capabilities.
- **Phase 3: Shift-Left Validation** - Integrating continuous testing and linting from the first commit to ensure code quality and prevent technical debt.
- **Phase 4: Work Package Distribution** - Execution of five specific work packages: WP1 (Core Infrastructure), WP2 (Secure Auth), WP3 (Dynamic Storefront), WP4 (AI & Logistics), and WP5 (User Acceptance).
- **Phase 5: Logistics and Notification Pipeline** - Finalizing the real-time notification system for order status transitions.

5.2 Risk Management

Risk ID	Identified Risk	Mitigation Strategy	Impact
R1	AI Query Hallucination	Prompt engineering and context-window constraints.	Medium
R2	Multi-campus Latency	CDN edge caching and optimized TTI benchmarks (1.2s).	High
R3	Firebase Data Scale	Implementing optimized Firestore indexing and sharding.	Low
R4	Authentication Bypass	Multi-stage OTP verification and secure AuthContext.	Critical

Table 1: Project Risk Assessment and Mitigation Matrix

5.3 Software Requirement Specification (SRS)

Functional Requirements (FR):

1. User registration with university email verification.
2. Secure 6-digit OTP-based authentication flow.
3. Dynamic product catalog with category-based filtering.
4. Real-time inventory tracking and stock status display.
5. AI-powered shopping assistant for product inquiries.
6. "Design Your Own" custom merchandise preview module.
7. Persistent shopping cart management and state recovery.
8. Real-time order status tracking and logistics notifications.
9. Administrative CMS panel for product and user management.

10. Multi-campus logistics routing for Etlik, Bilkent, Esenboğa.

Non-Functional Requirements (NFR):

1. **Performance:** Time-to-Interactive (TTI) must be less than 1.2s on campus networks.
2. **Scalability:** Support up to 5,000 concurrent student users during peak periods.
3. **Security:** SSL encryption for all data in transit and Firebase Security Rules for DB.
4. **Responsiveness:** Fully adaptive UI for mobile, tablet, and desktop viewports.
5. **Availability:** 99.9% system uptime via serverless Vercel/Firebase infrastructure.

6 Team, Resource Plan & Cost Estimation

6.1 Git-Based Contribution Audit

Based on the audit of the `muhammed_latest_patch` branch, the following roles are assigned:

- **Muhammed Yıldız (Core Architect):** Established project infrastructure, Firebase Auth, CI/CD pipelines, and core state management (AuthContext).
- **Utkan Dindaroğlu (Scientific Research):** Led the modular refactoring, advanced UX animations, and HL-CLDE ecosystem design.
- **Ahmet Selim Yılmaz (Process Design):** Defined the 5-phase methodology and SRS alignment with university standards.
- **Seyfullah Gülyazı & S. Fatih Sert (Implementation):** Managed specialized UI components and the logistics notification engine integration.

6.2 COCOMO Cost Estimation

Calculated using the **Organic Mode** for a team-driven academic project:

- **Estimated KLOC:** 2.69 (Actual repository audit).
- **Effort Equation:** $E = 2.4 \times (KLOC)^{1.05}$.
- **Result:** $E \approx 6.8$ Person-Months.
- **Schedule:** $T = 2.5 \times (E)^{0.38} \approx 5.2$ Months for total project completion.

7 Conclusion

AYBUStore provides a scalable and intelligent solution for campus commerce, leveraging modern web technologies to enhance the university's digital brand.